PROGRAM 12 – TIC TAC TOE GAME

AIM

To write a Python program to implement the Tic Tac Toe game between two players using a 3 × 3 game board.

ALGORITHM

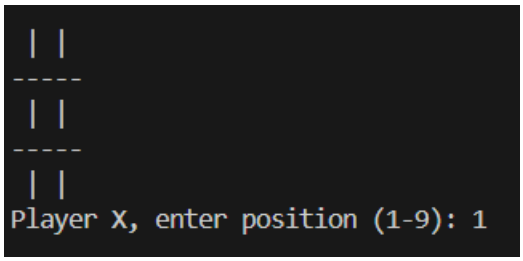
1. Start the program.
2. Create a 3 × 3 empty game board.
3. Assign X to Player 1 and O to Player 2.
4. Display the game board.
5. Read the position from the current player.
6. Check whether the selected position is empty.
7. Place the player's symbol in the selected position.
8. Check whether a player has won.
9. If all positions are filled without a winner, declare a draw.
10. Otherwise, switch to the next player.
11. Display the final result.

CODE

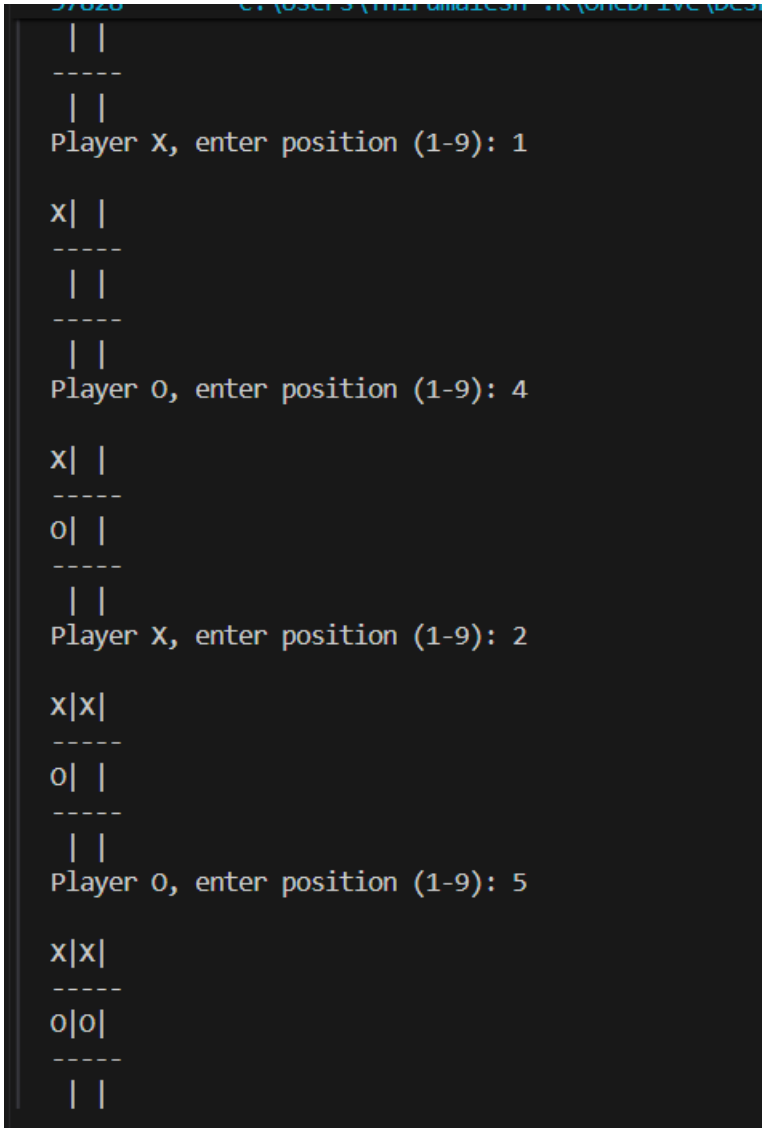
```
board=[" "]*9
def display():
    print()
    for i in range(0,9,3):
        print("|".join(board[i:i+3]))
        if i<6:
            print("-----")
def check_win(player):
    wins=[(0,1,2),(3,4,5),(6,7,8),(0,3,6),(1,4,7),(2,5,8),(0,4,8),(2,4,6)]
    return any(board[a]==board[b]==board[c]==player for a,b,c in wins)
player="X"
for turn in range(9):
    display()
    try:
        pos=int(input("Player "+player+", enter position (1-9): "))-1
    except ValueError:
```

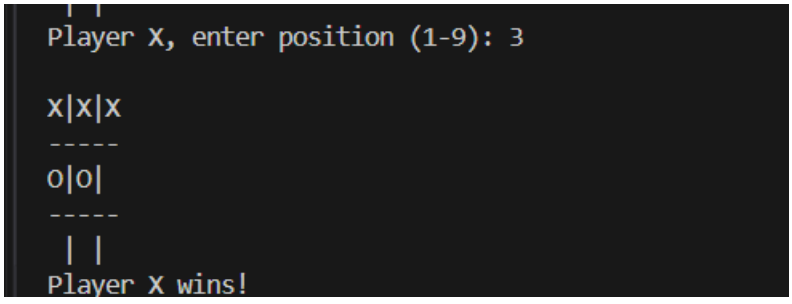
```
            print("Enter a valid number")
            continue
        if pos<0 or pos>8 or board[pos]!=" ":
            print("Invalid position, try again")
            continue
        board[pos]=player
        if check_win(player):
            display()
            print("Player",player,"wins!")
            break
        player="O" if player=="X" else "X"
    else:
        display()
        print("Game Draw!")
```

INPUT



OUTPUT





RESULT

Thus, the Tic Tac Toe game was successfully implemented in Python using a 3 × 3 game board.